LITTLE HAUNTS

3D Browser Game — Product & Technical Specification

Version: 0.3 — Approved direction with owner corrections **Status:** Creative and technical direction approved (v0.2 review). v0.3 corrections awaiting owner confirmation. No implementation started. **Platform:** Modern desktop browsers first; mobile adaptation later **Tagline:** *Small monsters. Big trouble.*

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0. Changelog

v0.3 (2026-10-10), owner corrections to v0.2

- **Primary art references registered:** the concept board (REF-00) and the Wick production character sheet (REF-01), both pending the owner's final approval. Wick's description is brought into line with REF-01:
  - The star is a stitched fabric patch; the clasp is a separate brass button.
  - Knit mitten hands.
  - About 3.5 heads tall and about 0.6× human height (≈1.05 m), so his collider radius is now 0.38 m.
  - Lantern in his left hand.
  - Curious and Determined expressions added; the drift clip is now `move_glide`.
  - Palette regrouped by element (§3, §4, §11.2).
- **Bone ownership and conservation:**
  - One source of truth for where each bone is.
  - An explicit list of allowed moves.
  - Invariants checked every tick.
  - A defined rule for rebuilding bone positions on restore.
  - Load-time validation and a new `CP_BONES_COMPLETE` checkpoint (§7.7, §9.2.1, §10.5).

- **Reveal state gates interaction:** reveal state is now part of interaction-target eligibility, evaluated in a fixed order every tick. Unrevealed objects have no prompt, can't be targeted, and can't be interacted with (§7.2, §7.3).
- **Spike extended to 3 working days.** It's feasibility-only and separate from final asset production, and spike assets can never ship (§12, §16).
- **Swept-circle collision** against segment and circle primitives replaces the speed assertion. Trigger crossing is also swept, and a thin-geometry test gauntlet is added (§7.1, §11.3, §14.1).

v0.2 (2026-10-10)

- **Wick is now a floating ghost.** Boots and legs removed. Added a motion spec for hover, sway, lean, and the tail wisp, an expression set, and a rig budget (§4.1–4.2, §11.2).
- **Rapier removed from the MVP.** Replaced with deterministic 2D collision on the ground plane. Colliders are authored in Blender and extracted at build time (§7.1, §11.3).
- **Added a disposable, timeboxed technical spike (Phase 1a).** It runs before architecture and estimates are finalized (§12, §16).
- **Deterministic gameplay is enforced by automated checks:** dependency rules, lint rules, a DOM-free tsconfig, a restricted math surface, replay-hash tests, and owner-held holdout tests (§10.3–10.4).
- **Puzzle, NPC, secret, checkpoint, and reset behavior are fully defined as state machines.** Typed conditions and actions replace `requirements: string[]` (§9, §10.5).
- **Acceptance criteria split into four kinds:** automated, measured on hardware, structured playtest, and human art review (§14).
- **Controls and camera corrected.** The camera is a fixed isometric view and movement is screen-relative. The lantern is held to raise, with a toggle option. Dialogue has advance keys, and foreground objects fade when they block the view (§5).
- **Added a visual reference register.** REF-00 is registered; the original screenshots are pending from the owner. No substitute references may be used (§3).
- **Open Decisions 1, 2, 3, and 7 closed.** New open decisions added (§20).

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1. Executive Summary

Little Haunts is a whimsical, atmospheric, isometric 3D adventure. It stars Wick, a timid little hooded ghost who floats through the world carrying a magical brass lantern. Players explore

handcrafted miniature dioramas, solve environmental puzzles, meet eccentric supernatural creatures, and collect lost souls. The tone is cozy Halloween charm, mystery, and physical comedy, not horror.

MVP objective: a polished, playable vertical slice of 5–10 minutes, *The Forgotten Graveyard*, accessible at a public browser URL. It must be a real interactive game, not a cinematic or pre-rendered mockup. It must also deliver the premium 3D miniature-diorama look established in the references (§3) at runtime, not only in screenshots.

2. Audience & Experience Goals

- Accessible to casual players with keyboard and mouse; no prior gaming expertise required.
- Visually distinctive, delightful, and readable at common desktop resolutions.
- Rewards curiosity with reactions, secrets, collectibles, and humorous encounters.
- Teaches mechanics through play with minimal text.
- Keeps one consistent, original visual identity across characters, props, and environments.

Non-goals for the MVP

Multiplayer, procedural open worlds, live generative-AI NPCs, accounts, commerce, backend persistence, mobile-first controls, camera rotation, click-to-move, gamepad support, control remapping, a physics engine, and a full five-level campaign.

3. Visual References

References define the creative target. They are **inspiration, not source assets**. Shipped assets must be original (§11.4).

ID	Reference	Role	Status
REF-00	Little Haunts concept board: title art, Wick sheet, environment thumbnails (Graveyard, Ravenhold Keep, Whispering Woods, Clockwork Crypt), NPC and creature lineup, props, palette and lighting. File: `49486092-B26F-4B03-B765-C8E4C09E0CA1.png`	**Primary.** World, environments, NPCs, props, palette, lighting, overall tone	Registered 2026-10-10. **Pending owner final approval.**

REF-01	Wick production character sheet: hero render, 5-view turnaround, detail callouts, lantern turnaround, expressions, action poses, scale and proportions, palette, key visual. File: `d0082a463508a5a3120136ff4c96756669eafe7e.png`	**Primary.** Wick and his lantern	Registered 2026-10-10. **Pending owner final approval.**

Rules

- **Use before approval:** REF-00 and REF-01 guide all art work now. Until the owner gives final approval, the art bible stays draft and no asset can be marked `accepted` against them. Final approval (date and owner) is recorded in this table and is required for Gate 2.
- **Precedence:** REF-01 wins over REF-00 for Wick and the lantern. REF-00 governs everything else. References added later take precedence only where the owner says so.
- **Who adds references:** only the owner. Agents and contributors must not generate, search for, or substitute references. A missing reference is recorded as a blocking question.
- **Inconsistencies:** if the references disagree with each other or with themselves, the owner decides and the decision is logged here. For example: lantern in the left hand, per REF-01's front views (adopted; awaiting confirmation).

4. Art Direction Bible

4.1 Character: Wick

All details per REF-01.

- **Silhouette and proportions:** a small, toy-like figure about 3.5 heads tall overall, roughly 0.6× the height of an adult human. The oversized pointed hood flops toward the back.
- **Cloak:** deep navy-to-indigo knit or felt fabric with visible running-stitch seams and faint warm flecks. The hood edge has a soft, fuzzy fiber halo.
- **Star patch:** a tan, stitched fabric star on the chest. It's a patch, not a clasp or a metal emblem.
- **Cloak clasp:** a small, round brass button at the collar.
- **Hands:** small knit mittens emerging from the sleeves, with no fingers.
- **Face:** smooth, pale, porcelain-like, with large glossy dark eyes. Expressions come

mainly from eye shape, plus small brow and blush cues.

- **Ghost form: no legs or boots.** Below the cloak hem is a translucent, pale-cyan wisp tail that trails and curls.
- **Lantern:** a stylized, slightly worn brass lantern held in Wick's **left** hand (per the front and front-3/4 views). Its warm, flickering amber light is the brightest warm accent on screen.
- **Readability:** must read clearly at isometric gameplay distance (measurable target in §14.2).
- **Personality:** "nervous, brave, and always getting into trouble." He startles easily but pushes forward anyway (the Determined expression), and celebrates small victories with big joy.

4.2 Wick motion and animation

Wick's motion has two layers.

The **procedural layer** runs every frame in the presentation layer. It never affects gameplay state or collision. All values are initial tuning values, adjustable in config.

Motion	Behavior
Hover bob	Continuous vertical bob, amplitude ≈ 0.05 m, period ≈ 2.2 s. Damped while moving fast.
Cloak sway	Spring-damper on cloak bones, driven by acceleration and turning. Lags behind body motion.
Hood tip	Secondary spring motion; flops on stops and turns.
Lean	Pitches toward the direction of travel (up to ≈ 12°) and rolls into turns (up to ≈ 8°). Eases back at rest.
Tail wisp	Bone chain or shader trail following recent positions. Stretches during dash, curls at idle.
Stop settle	Brief overshoot-and-settle when he decelerates to a stop.
Lantern swing	Pendulum motion on the lantern from hand velocity. Steadies while the lantern is raised.

The **authored clip layer** blends on top. Expressions and action poses match REF-01.

Expressions: Neutral, Curious, Happy, Surprised, Scared, Determined. Action poses: Idle Hover, Move/Glide, Look, Interact, Lantern Raise.

Clip	Use
`idle_hover`	Breathing, small arm and lantern adjustments
`move_glide`	Forward-leaning glide posture with the tail streaming behind (body motion comes from the procedural layer)
`move_dash`	Streamlined, tucked pose; tail extended
`look`	Head and eye turns toward points of interest, lantern held out; used by scripted beats
`curious`	Head tilt and lean-in when near an unrevealed revealable or a new interactable
`interact`	Reach or push toward the target, lantern lowered toward it (per REF-01)
`lantern_raise` / `lantern_hold` / `lantern_lower`	Raise, sustained hold loop, lower
`surprised`	Jump-back recoil with wide eyes
`scared`	Mittens over eyes, cower and tremble (per REF-01); used in comedic beats
`determined`	Brief squared-up, lantern-forward beat before entering a puzzle area or after a reset
`happy` / `celebrate`	Bounce or spin; used on puzzle solves and collectibles
`shrug`	Blocked or failed interaction feedback

Blending rules: crossfade over 0.15–0.25 s between clips; the procedural layer is applied additively after clips; no visible pops. Facial expressions (neutral, blink, curious, wide/surprised, happy squint, scared, determined, look-left, look-right) use either a texture atlas or morph targets; the spike decides which (§12).

4.3 World and materials

- Handcrafted diorama proportions: exaggerated architectural detail, worn chunky

stonework, twisted trees, carved tombstones, moss, fabric, aged brass, weathered wood.

- Surfaces read as premium miniatures: beveled edges, tactile texture, crafted imperfection.
- **Avoid:** photorealism, mismatched asset styles, flat or untextured placeholder geometry in the final build, excessive bloom, and scenes too dark to read.

4.4 Palette and lighting

- **Palette** (swatch groups per REF-01; scene-wide accents per REF-00):
  - **Cloak:** deep navy, indigo-slate, pale gray.
  - **Star and accents:** cream, gold-tan, warm brown.
  - **Ghost:** near-white, pale cyan.
  - **Lantern:** pumpkin orange, amber, warm cream.
  - **Environment:** midnight navy, slate blue, stone gray, moss green, earthy brown.
  - **Scene accents from REF-00:** dusty purple and muted turquoise (glowing mushrooms, magic).
  - Exact hex values are sampled from the registered references in Phase 0; agents do not invent them.
- **Lighting:** strong warm/cool contrast. Cool moonlit ambient light; warm pools from candles, lanterns, and jack-o'-lanterns. The environment uses mostly baked static lighting. Realtime light is reserved for Wick's lantern and puzzle effects. Soft shadows; restrained bloom.
- **Atmosphere:** layered ground fog, floating embers and dust, subtle vegetation motion, distant silhouettes for depth.

4.5 MVP content scope from REF-00

- **NPCs:** the skeleton (Gatekeeper) is the only MVP NPC. The pumpkin creature, blue wisp ghost, beaked-mask figure, and black cat are future scope. A non-interactive ambient raven is a stretch goal.
- **Environments:** only The Forgotten Graveyard. Ravenhold Keep, Whispering Woods, Clockwork Crypt, and Midnight Carnival are future scope.
- **Props:** see the required set dressing in §8.3.

5. Camera & Controls

5.1 Camera

- **Orthographic, fixed isometric.** No player-controlled rotation or zoom in the MVP.
- Default yaw 45°. Default pitch 35.26° (true isometric). Pitch may be tuned between 30° and 40° and is locked at Gate 2.
- **Framing:** a fixed orthographic size per level, initially showing about 14–18 m of world width at 16:9. Final value set at Gate 2.
- **Follow:** a central dead zone of about 15% of the viewport, with critically damped smoothing of about 0.25 s. The camera is clamped to authored level bounds so the edges of the diorama frame the scene.
- **Scripted framing:** puzzle reveals and NPC beats may move the camera for at most 2 s before handing control back. The camera must never hide an active puzzle element.
- **Occluder fade:** geometry tagged `occluder` that comes between the camera and Wick, or the current interaction target, fades to about 25% opacity within 0.2 s using dithered transparency.

5.2 Movement mapping

Movement is **screen-relative**: W or Up moves Wick toward the top of the screen. Concretely, that means along the camera's forward vector projected onto the ground plane and normalized. A/D and S follow the same rule. Diagonal input is normalized so diagonals are not faster. Opposite keys held together cancel out.

5.3 Controls (desktop, keyboard-only MVP)

Input	Action
WASD / Arrow keys	Move (screen-relative)
Shift (hold)	Dash. Ignored while the lantern is raised.
E	Interact with the highlighted target
Space (hold)	Raise lantern. Releasing lowers it. A hold/toggle setting is available.
	Advance dialogue (while dialogue is shown, these keys don't

E / Space / Enter	move or raise the lantern)
Esc	Open or close pause. During dialogue, pauses without skipping.
Arrow keys / Tab / Shift+Tab, Enter	Navigate and select in menus
Mouse	Menu interaction only; no camera or movement control

- All controls are shown on first use (movement, interact, lantern) and are always listed in Pause → Controls.
- Keyboard focus is trapped in open menus and returned to the game on close.
- Input is converted to abstract **intents** (move vector, dash, interact, lantern, advance, pause). Gameplay never reads keys directly, so click-to-move, gamepad, and remapping can be added later without touching gameplay.

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6. Core Game Loop

Explore → Notice → Interact → Solve → Discover → Unlock

1. Float through a compact isometric diorama.
2. Raise the lantern to reveal hidden runes, objects, and passages.
3. Interact with NPCs and physical props.
4. Solve a sequence of environmental puzzles.
5. Find optional secrets and collectibles.
6. Unlock the exit and finish the level.

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7. Mechanics

All gameplay runs on a **fixed 60 Hz simulation tick** (§10.3). Durations below are in milliseconds and are converted to whole ticks by rounding up. Scale is 1 world unit = 1 m. Numbers are initial tuning values unless marked otherwise.

7.1 Movement and collision (no physics engine)

- **Movement:** drift at up to 2.4 m/s; dash at up to 4.2 m/s. Wick reaches full speed in

about 120 ms and stops in about 180 ms. With the lantern raised, speed is multiplied by 0.6.

- **Collision is 2D on the ground plane and swept (continuous).** Wick is a circle with radius 0.38 m, scaled to his 1.05 m height.
- **Primitives:** at extraction (§11.3), every static collider compiles to two primitive types:
  - **Segments:** box and rotated-box edges, curve chains, and zero-thickness fence or railing lines.
  - **Circles:** posts and cylinders.

  Sweeping Wick's circle (radius r) against a segment is a ray test against a capsule of radius r. Against a circle of radius c, it's a ray test against a circle of radius r + c. The capsule's rounded ends cover segment endpoints, so corners need no special case.
- **Each tick:**
  1. Sweep the desired displacement against nearby primitives (found with a uniform grid of 2 m cells) to get the earliest time of impact.
  2. Advance to that point, minus a 1 mm skin.
  3. Project the remaining displacement onto the contact surface (slide) and sweep again.
  4. Stop after 3 sweeps and discard any remainder, so corners stop cleanly instead of jittering.
  5. Run a final depenetration pass as a numerical safety net. It should never do anything in normal play; dev builds count and log each time it acts.
- **Why it handles thin geometry:** the sweep tests the whole path, so a zero-thickness fence blocks Wick at any speed, after any frame hitch, and during any scripted move. Correctness doesn't depend on speed × tick time staying under a threshold. Scripted moves of Wick use the same sweep, or target positions validated in level data.
- **Determinism:** time-of-impact math uses only the allowed operations (§10.3, including `Math.sqrt`). Contacts at equal times are resolved in primitive-ID order.
- **Height is presentation-only.** Wick's on-screen height is the baked ground height at his position plus the hover offset. Floating means uneven terrain needs no foot placement or slope handling. Ledges and walls are simply colliders.
- **Triggers:** 2D volumes that emit enter and exit events into gameplay state. Entry is tested against the swept path of Wick's center, not only his final position, so a dash

can’t skip a thin trigger such as the exit threshold.

- **Debug overlay** (dev builds only): F1 draws colliders, triggers, interaction ranges, and the lantern radius.
- **When to revisit:** add a physics library only if an approved puzzle needs moving rigid bodies. That would require an amendment to this spec.

7.2 Lantern reveal

- **Hold Space** (or toggle, per settings) to raise. Over 250 ms the reveal radius grows from 0 to its full 3.0 m. Releasing lowers the lantern and shrinks the radius over 150 ms.
- An object counts as **lit** when its anchor point is inside the current radius.
- An object is **revealed** after it has been lit continuously for its dwell time (default 300 ms).
- Each revealable object has one of two modes:
  - `persistOnReveal` : stays revealed permanently once revealed; saved immediately (§7.7).
  - `whileLit` : revealed only while lit, with a 200 ms grace period at the radius edge so it doesn’t flicker.
- **Reveal state gates interaction.** An object that isn’t currently revealed has no highlight and no prompt, and it can’t be targeted (§7.3). For `whileLit` objects, the moment reveal state drops (after the grace period), the object stops being eligible in that same tick. An interact press on that tick or any later tick does nothing to it.
- Revealing emits presentation cues (sound, light pulse, materialize effect). Reveal logic is purely deterministic distance and time math; no AI inference.
- The lantern cannot be raised during dialogue or scripted beats.

7.3 Interactions

- Each interactable has a range (default 1.2 m from Wick’s center), a prompt, a condition for when it can be targeted, and the actions it triggers (§10.5).
- **Eligibility is evaluated every tick, and it’s the only path to an interaction.** An interactable is eligible when Wick is in range, its `targetableWhen` condition holds, and, if it’s attached to a revealable, that revealable is currently revealed. The targeting system applies the reveal requirement itself; level authors don’t write it, so it can’t be forgotten. Level validation rejects any attempt to override it.

- **Fixed order within each tick:**

  1. Movement and collision.
  2. Lantern radius and reveal state.
  3. Eligibility and target selection.
  4. The interact intent, applied only to the target selected in step 3.
  5. Puzzle and NPC state machines.
  6. Checkpoint writes.

  With no eligible target, an interact press does nothing; it's not a blocked interaction, because nothing was targeted. A target that becomes ineligible is dropped on the same tick.

- **Exactly one target at a time.** Candidates in range are scored by distance and by how closely they line up with Wick's facing. The highest score wins.
- To prevent flicker, the target only switches when a new candidate scores at least 15% higher.
- The target gets a highlight rim, and the prompt shows the key glyph and verb (for example "**E** Push").
- **Every interaction produces feedback.** A successful interaction plays its outcome cue. A blocked one plays a `shrug` animation plus a short fizzle sound. Nothing is silent.

7.4 Dialogue and scripted beats

- Dialogue is authored and shown one line at a time in a speech bubble above the speaker, advanced with E, Space, or Enter.
- Holding a key never auto-advances. Lines are short (at most about 80 characters each).
- Scripted beats lock player control, last at most 8 s each, and always return control. They can't be skipped in the MVP, which is why they're kept short.
- Beats are sequences of data-defined steps (camera framing, NPC state change, cue, wait) run by the gameplay layer. The presentation layer plays them back.

7.5 Puzzle framework

- Every puzzle is a **data-defined finite state machine**: states, events, guarded transitions, and actions (§10.5).

- Each state maps to one of four statuses: `locked`, `available`, `inProgress`, or `solved`.
- Transient states (for example, a reset animation) must have a fixed duration of at most 2 s and leave automatically.
- Every puzzle defines what happens on reload and on Restart Level (§9.4).
- Puzzles and NPCs talk only through events and conditions. No puzzle reads another puzzle's internal state.

7.6 Collectibles and secrets

- The MVP has one lost soul collectible (`LOST_SOUL_01`) and one secret passage (§9.3).
- Optional content never gates the main path, and no main-path item is placed in optional areas.
- Collectibles are unique per save. Once collected, they stay collected across Restart Level.

7.7 Saves and checkpoints

- Progress is stored in versioned `localStorage` keys: `littlehaunts.save.v1` for game state and `littlehaunts.settings.v1` for settings.
- **Checkpoints** capture the main path. On load, the game rebuilds that checkpoint's canonical state, then layers immediately-saved data on top. Any puzzle progress made after the checkpoint is discarded.

Checkpoint	Written when	Canonical state on load
`CP_START`	Player first gains control	Start of level; lantern not yet acquired
`CP_LANTERN`	Lantern picked up	Lantern held; Three Lanterns `Ready`
`CP_LAMPS_SOLVED`	Three Lanterns solved	All lamps lit; courtyard gate open; skeleton `Idle`
`CP_BONES_STARTED`	Skeleton enters `AwaitingHelp`	Skeleton `AwaitingHelp`; all bones at their spawn points; nothing carried
`CP_BONE_DELIVERED`	Each delivery that leaves bones outstanding	Skeleton `AwaitingHelp`; bone locations rebuilt by the restore rule in §9.2.1
	The final delivery	All bones delivered; skeleton

`CP_BONES_COMPLETE`	(skeleton enters `Reassembling`)	`Reassembling` (the gag replays)
`CP_GATE_OPEN`	Skeleton `Thankful` dialogue ends	Exit gate open; skeleton `Done`

- **Saves are atomic.** Each write serializes the complete save object to one key, in the same tick that the triggering state change is committed. There are no partial or incremental writes.
- **Saved immediately, outside checkpoints:** collectibles, `persistOnReveal` reveals, the secret-passage-open flag, settings, and elapsed play time (also saved on pause and on `visibilitychange`).
- **Loading:** unknown or newer schema versions, invalid JSON, or data that fails validation fall back to a new game without crashing. The raw data is backed up to `littlehaunts.save.corrupt.<timestamp>` and a warning is logged. Older schemas are migrated by pure, tested functions.
- **Restart Level** (in the pause menu) returns to `CP_START` with fresh puzzle and reveal state and a new timer. It keeps collectibles, the best completion record, and settings.
- **New Game** (title screen, with confirmation) wipes everything except settings.
- **Completion record:** play time counts from `CP_START` until Wick crosses the exit threshold. It excludes time paused, time loading, and time with the tab hidden. The completion screen shows time (mm:ss), lost souls found (x/1), and the best time.

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8. First Level — The Forgotten Graveyard

Target length: 5–10 minutes on a first playthrough (measured in §14.3).

Opening beat: Wick awakens in the hollow. A voice says, *"The dead have gone missing."* A skeleton bursts out of a nearby grave. Wick and the skeleton startle each other and flee in opposite directions. The skeleton squeezes between the bars of the locked courtyard gate, because being made of bones has its advantages.

8.1 Zones and gating (provisional layout; finalized in Phase 0)

Zone	Contents	Gated by
Z1 Waking	Spawn point, intro beat, lantern pickup, first	—

Hollow	rune (Rune 1)	
Z2 Rune Row	Rows of tombstones, Runes 2 and 3 (hidden), cracked wall to the secret alcove	—
Z3 Lamp Circle	Three stone lamps around a plinth; Bone A	—
Z4 Mausoleum Courtyard	Skeleton; Bones B and C	Courtyard gate (opened by solving Three Lanterns)
Z5 Secret Alcove	`LOST_SOUL_01`	Secret passage (optional)
Z6 Graveyard Gate	Exit threshold, end beat	Exit gate (opened by the Gatekeeper)

8.2 Level beats

1. **Arrival:** intro beat; movement prompt shown; Wick picks up the lantern (first **E** prompt).
2. **First reveal:** a raise-lantern prompt appears near Rune 1. Raising the lantern reveals it, teaching both the reveal and the “runes mean something” rule.
3. **Lantern puzzle:** find Runes 2 and 3, then light the three lamps in the correct order (§9.1). The courtyard gate opens.
4. **Skeleton encounter:** Wick startles the skeleton again, and this time it falls apart. Wick retrieves three scattered bones, then watches a botched reassembly gag (§9.2). The exit gate opens.
5. **Secret detour (optional):** the lantern reveals a cracked wall in Rune Row; behind it is the lost soul (§9.3). Available from beat 2 until the level ends.
6. **Exit:** Wick floats through the graveyard gate. A short celebration plays, then a tease of the next area, then the completion screen.

8.3 Required set dressing

Ruined stone walls and arches, varied carved gravestones, a wrought-iron gate and fence runs, a twisted tree, moss, glowing mushrooms, pumpkins (including jack-o′-lanterns), candles and a candelabra, lantern props, crosses, distant silhouettes of a manor or keep, ambient fog, and subtle environmental animation (flickering flames, swaying branches, drifting embers).

9. Puzzle & NPC State Machines

These machines are normative. Implementations must match them, and tests (§14.1) cover every transition listed. Event names are illustrative; the IDs in the final level data must map one-to-one onto these.

9.1 The Three Lanterns (`PZ_THREE_LANTERNS`)

Elements

- Runes R1, R2, R3: `persistOnReveal` revealables. Each shows a glyph (Moon, Raven, Key).
- Lamps L_MOON, L_RAVEN, L_KEY: interactables, each carved with its glyph.
- **Order source** (provisional; finalized in the GDD): each rune carries 1–3 carved notches, and the order is ascending notch count. The order is authored in level data as `order: [LampId, LampId, LampId]` .

Context: `k ∈ {0,1,2,3}` is the number of lamps correctly lit so far.

Targeting: lamps become targetable once the lantern is acquired. A lamp that is already lit can't be targeted, so the player can't trigger a reset by accident.

```
stateDiagram-v2
  [*] --> Locked
  Locked --> Ready: LANTERN_ACQUIRED
  Ready --> Ready: LAMP(x) [rune x hidden] / fizzle
  Ready --> Ready: LAMP(x) [rune x revealed, x != order 0] / wrongPuff
  Ready --> Sequencing: LAMP(x) [rune x revealed, x == order 0] / light x, k=1
  Sequencing --> Sequencing: LAMP(x) [rune x hidden] / fizzle
  Sequencing --> Sequencing: LAMP(x) [revealed, x == order k, k < 2] / light x, k
  Sequencing --> Solved: LAMP(x) [revealed, x == order k, k == 2] / light x, k=3
  Sequencing --> Resetting: LAMP(x) [revealed, x != order k] / extinguishAll
  Resetting --> Ready: after 1000 ms / k=0
  Solved --> [*]
```

State	Status	Entry actions
Locked	locked	Lamps not targetable
Ready	available	k = 0; all lamps unlit
Sequencing	inProgress	—

Resetting	inProgress (transient, 1000 ms)	Comedic puff of smoke from every lit lamp; Wick plays `scared` ; lamp input ignored
Solved	solved	Success cue; Wick plays `celebrate` ; scripted camera framing on the courtyard gate (≤ 2 s); gate opens; write `CP_LAMPS_SOLVED`

Rules

- Interacting with a lamp whose rune is still hidden never resets progress. Curiosity is never punished.
- Runes stay revealed through resets and reloads. Re-solving after a mistake never requires rediscovery.
- **Reload mid-sequence:** returns to `Ready` with k = 0. **Restart Level:** returns to `Locked` .
- **No soft-lock:** lamps are always re-targetable outside `Resetting` , and `Resetting` always exits. Brute-forcing the order is possible (6 permutations), and that's acceptable.

9.2 The Gatekeeper's Bones (`PZ_GATEKEEPER_BONES`) and the skeleton NPC (`NPC_GATEKEEPER`)

Elements

- Skeleton NPC (the Gatekeeper).
- Bones (`ItemId` s): Bone A, the left arm, visible on the path in Z3. Bone B, the right leg, a `persistOnReveal` revealable hidden in fog in Z4. Bone C, the ribcage, on a mausoleum ledge in Z4 reached by a short ramp detour.
- The exit gate.

Rules

- Bones are picked up with **E**. Wick can carry any number. Carried bones show as HUD icons.
- Talking to the skeleton (**E**) delivers every bone Wick is carrying.
- No bone may be placed in optional areas. Bones can't be dropped or lost.

```
stateDiagram-v2
  [*] --> Buried
  Buried --> Rising: INTRO_TRIGGER
```

```
Rising --> StartledIntro: rise complete
StartledIntro --> Fleeing: after 1200 ms
Fleeing --> Idle: reached courtyard
Idle --> Startled: WICK_ENTERS_COURTYARD
Startled --> Scattered: after 800 ms
Scattered --> AwaitingHelp: bones landed (1500 ms) / write CP_BONES_STARTED
AwaitingHelp --> AwaitingHelp: TALK [carrying 0] / next hint
AwaitingHelp --> AwaitingHelp: TALK [carrying >= 1, delivered + carrying < 3] /
AwaitingHelp --> Reassembling: TALK [delivered + carrying == 3] / deliver, writ
Reassembling --> Reassembled: gag sequence (<= 6 s)
Reassembled --> Thankful: after 500 ms
Thankful --> Done: dialogue ends / open exit gate, write CP_GATE_OPEN
Done --> [*]
```

Skeleton state	**Puzzle status**	**Behavior**
Buried → Rising → StartledIntro → Fleeing	locked	Intro beat. Wick plays `surprised`, then a scripted recoil. The skeleton flees through the courtyard gate bars.
Idle	locked	Ambient idle in Z4; Z4 is unreachable until Three Lanterns is solved
Startled	inProgress (transient)	The skeleton jumps, and Wick jumps too
Scattered	inProgress (transient)	The skeleton collapses; bones arc to their spawn points; the skull lands upright
AwaitingHelp	inProgress	Skull dialogue. Hints cycle in a fixed order, each pointing toward the nearest bone that is neither delivered nor carried. Bone B's hint mentions "something glinting in the fog" to cue the lantern.
Reassembling	inProgress (transient)	Botched-reassembly gag (for example, the skull goes on backwards, then gets corrected); Wick plays `happy`
Reassembled → Thankful	solved	Thank-you dialogue; scripted framing on the exit gate (≤ 2 s)
Done	solved	The skeleton waves near the open exit gate

Rules

- **Reload:** restores the checkpoint (§7.7) using the bone restore rule in §9.2.1.
- **Restart Level:** returns to `Buried`, and the intro beat replays.
- The skeleton can't re-scatter once it reaches `AwaitingHelp`.
- **No soft-lock:** every bone is reachable once Z4 opens, bones can't be lost, and hints always point to one that remains.

9.2.1 Bone ownership and conservation

Single source of truth. Game state holds exactly one structure for bones, `boneLocation: Record<BoneId, BoneLocation>`, covering bones A, B, and C. There's no separate inventory list or counter. Carried bones, delivered bones, HUD icons, hint targets, and the skeleton's progress are all derived from this map.

Location	Meaning	World pickup exists?
`onSkeleton`	Before the scatter	No
`atSpawn`	Lying at its authored spawn point	Yes. Bone B is eligible only once revealed (§7.3).
`carried`	Held by Wick	No
`delivered`	Returned to the skeleton	No

Allowed moves. These are the only code paths that change the map:

- `onSkeleton → atSpawn`: all three bones together, on entering `Scattered`.
- `atSpawn → carried`: a pickup interaction. The pickup is eligible only while that bone is `atSpawn`.
- `carried → delivered`: TALK in `AwaitingHelp`. Every carried bone moves in the same tick.
- `carried → atSpawn`: during checkpoint restore only.

Nothing moves a bone out of `delivered`, and nothing creates or removes a bone.

Invariants. These are checked after every tick in dev and test builds, where a violation fails the test. In production, a violation logs an error and restores the last checkpoint.

1. Exactly three bone IDs exist, and each has exactly one location.

2. A world pickup exists for a bone if and only if that bone is `atSpawn`.
3. Bone locations match the skeleton's state:
    - Skeleton `Buried` through `Startled`: every bone is `onSkeleton`.
    - `Scattered` or `AwaitingHelp`: no bone is `onSkeleton`, and fewer than three are `delivered`.
    - `Reassembling` and after: every bone is `delivered`.
4. The HUD shows exactly the bones that are `carried`.

Save and restore

- Saves store only the set of delivered IDs (`bonesDelivered`). Carried bones are never saved.
- Restore rebuilds the map from the checkpoint:
    - Before `CP_BONES_STARTED`: every bone is `onSkeleton`.
    - From `CP_BONES_STARTED` on: IDs in `bonesDelivered` are `delivered`, and every other bone is `atSpawn`.
    - At `CP_BONES_COMPLETE` and after: every bone is `delivered`.

  So a carried bone returns to its spawn point exactly once, and a delivered bone stays delivered.
- **Load validation** rejects a save, falling back as described in §7.7, if `bonesDelivered` contains unknown or duplicate IDs, or if it contradicts the checkpoint. Examples: three bones delivered at `CP_BONE_DELIVERED`, or any bone delivered before `CP_BONES_STARTED`.
- **No in-between window:** delivery and its checkpoint write happen in the same tick (§7.7), so a reload can never observe a bone as both carried and delivered.

Tests: property-based tests run random intent sequences, including reloads and restarts at random ticks, and assert the invariants on every tick. The graph search (§9.5) also asserts them in every reachable state.

9.3 Secret passage (`SC_CRACKED_WALL`) and lost soul (`LOST_SOUL_01`)

```
stateDiagram-v2
  [*] --> Hidden
  Hidden --> Revealed: lit for dwell time (persistOnReveal)
```

```
Revealed --> Open: PUSH (E) / wall slides, set flag secret_open (saved immediat
Open --> [*]
```

- **The wall:** before it's revealed, it looks like ordinary wall and can't be targeted. Once revealed, it shows a glowing crack and the prompt "**E** Push".
- **The lost soul:** a faint shimmer is always visible; it is fully revealed (`whileLit`) and collectible only while lit. Collecting it plays `celebrate` and a chime, and it's saved immediately.
- If the lost soul was collected in an earlier run, it does not respawn after Restart Level. The completion screen still counts it (1/1).
- Wick can enter and leave the alcove freely, and no state change in it affects the main path.

9.4 Level-wide reset matrix

Event	Three Lanterns	Gatekeeper's Bones	Secret / lost soul	Reveals	Timer
Wrong lamp	→ Ready (k=0)	—	—	kept	continues
Reload	Checkpoint state	Checkpoint state	Kept (immediate save)	`persistOnReveal` kept	Restored from saved elapsed time
Restart Level	→ Locked	→ Buried	Wall → Hidden; collectible stays collected	Cleared	Reset
New Game	→ Locked	→ Buried	Everything reset	Cleared	Reset; best time cleared

9.5 Soft-lock freedom (provable)

The level's abstract state is small and finite: checkpoint, puzzle states, k, bone locations, reveals, secret state, and collectibles. CI runs a **breadth-first search** over that state space,

applying every enabled action (interact with each targetable object, reveal each revealable in an accessible zone, wait out transient states, reload, restart). The search must show:

1. From **every** reachable state, level completion is still reachable.
2. Every transient state exits within its fixed duration.
3. No reachable state outside a scripted beat has zero main-path actions available.
4. The bone invariants (§9.2.1) hold in every reachable state.

A separate **geometric reachability test** flood-fills a walkable grid built from the extracted collision data (0.1 m cells, inflated by Wick's radius). It checks, under every gate configuration, that every main-path interactable has a reachable cell within its interaction range.

---

10. Technical Architecture

The stack and layering below are **provisional until the spike (§12) is reviewed**. The determinism rules (§10.3–10.4) are fixed regardless of what the spike finds.

10.1 Stack

Layer	Choice	Notes
Rendering	Three.js / WebGL2	Orthographic camera, PBR materials, baked lightmaps, selective realtime lighting, post-processing
Language	TypeScript (strict)	Branded IDs; discriminated unions for all data contracts
Build	Vite	Static output
Collision	Custom 2D module in `src/game/collision`	Replaces Rapier (§7.1)
Assets	glTF 2.0 / GLB	Meshopt or Draco geometry compression; KTX2/Basis textures
Asset tooling	glTF-Transform (Node)	Validation, compression, collider and anchor extraction
	Three.js AnimationMixer	Procedural motion lives in the presentation layer

Animation	plus a procedural layer	only
Audio	Web Audio API	Audio context created or resumed only on the first user gesture
Persistence	Versioned `localStorage` behind a persistence port	Save encode/decode/migrate are pure functions in `src/game`
Hosting	Cloudflare Pages	Static; no server
Tests	Vitest + Playwright	Plus dependency-cruiser and ESLint architecture rules

No React unless a substantial UI need appears. The HUD and menus are plain DOM and CSS.

10.2 Layers

```
app/      boot, main loop (fixed-timestep accumulator + render interpolation), li
  │
  ├─► engine/   adapters: renderer, camera, asset loading, input → intents,
  │             audio, persistence port (localStorage), Wick procedural motion, c
  ├─► ui/       DOM HUD, prompts, dialogue bubbles, menus, completion screen
  └─► game/     PURE deterministic core: state, step(), intents, collision, lante
                interactions, puzzles, NPC machines, beats, checkpoints, save cod
      levels/   PURE level data (puzzles, interactables, dialogue, beats) + extra
```

- The core contract is `step(state, intents, tick) → { state, events }`. The game layer emits **events** (cues), and adapters render, animate, and play sounds in response. Adapters never change gameplay state.
- The renderer reads the latest two states and interpolates between them for smooth display.

10.3 Determinism rules (normative)

Code in `src/game/**` and `src/levels/**` must:

1. Import only from `src/game/**`, `src/levels/**`, or `src/shared/**` (pure types and utilities).
2. Never touch the platform: no `three`, no DOM, no Web Audio, no browser storage, no `window`, `document`, `navigator`, `performance`, `requestAnimationFrame`, `setTimeout`,

`setInterval`, `fetch`, or `Date`.

3. Never use `Math.random`. Randomness (for example, idle variation) comes from a seeded PRNG stored in the game state.
4. Advance only in fixed ticks of exactly 1/60 s. Time is a tick counter, never wall-clock time.
5. Use only correctly rounded arithmetic (`+ − × ÷`, `Math.sqrt`, `Math.abs`, `Math.min`, `Math.max`, `Math.floor`, `Math.round`) and precomputed constants. `Math.sin`, `cos`, `tan`, `atan2`, `exp`, `log`, and `pow` are banned: their results can differ in the last bits between browser engines. Angle math goes through `src/game/math`, which uses lookup tables or polynomial approximations built from the allowed operations.
6. Process collections in a stable order. Iterate arrays by ID order; never depend on the iteration order of a `Map` or `Set` built from unordered input.

10.4 Enforcement (CI fails on any violation)

Check	Mechanism
Layer boundaries	dependency-cruiser forbidden-path rules: `game` / `levels` → anything other than `game` / `levels` / `shared`; `engine` / `ui` / `app` → `game` internals, except through the public `game/index.ts`
Forbidden imports, globals, APIs	ESLint `no-restricted-imports`, `no-restricted-globals`, `no-restricted-properties` (the `Math.*` list and `Date.now`), and `no-restricted-syntax` (`new Date`), scoped to `src/game/**` and `src/levels/**`
No platform types	`tsconfig.game.json` with `lib: ["ES2022"]` and `types: []` (no DOM), so DOM usage fails typecheck
Runs headless	The game entry is imported and run in a Vitest **node** environment with no jsdom
Replay determinism	Recorded intent streams with fixed seeds, replayed 3× in Node and once each in Playwright Chromium, Firefox, and WebKit. The state hash must match exactly at every 60th tick and at the end.
Holdout acceptance tests	Owner-maintained tests stored outside the builder-accessible repository (for example, a private repo checked out only in CI). Builder agents can't read or modify them. Branch protection requires the holdout job to pass.

10.5 Data contracts (illustrative; finalized after the spike)

```
type Brand<T, B extends string> = T & { readonly __brand: B };
type EntityId = Brand<string, 'EntityId'>;
type PuzzleId = Brand<string, 'PuzzleId'>;
type ItemId = Brand<string, 'ItemId'>;
type FlagId = Brand<string, 'FlagId'>;
type CueId = Brand<string, 'CueId'>;
type DialogueId = Brand<string, 'DialogueId'>;
type PromptId = Brand<string, 'PromptId'>;
type CheckpointId = Brand<string, 'CheckpointId'>;
type BoneId = 'BONE_A' | 'BONE_B' | 'BONE_C';
type BoneLocation = 'onSkeleton' | 'atSpawn' | 'carried' | 'delivered';

type PuzzleStatus = 'locked' | 'available' | 'inProgress' | 'solved';

type Condition =
  | { kind: 'flag'; flag: FlagId; is: boolean }
  | { kind: 'puzzleStatus'; puzzle: PuzzleId; status: PuzzleStatus }
  | { kind: 'revealed'; entity: EntityId }
  | { kind: 'hasItem'; item: ItemId; min?: number }
  | { kind: 'all'; of: readonly Condition[] }
  | { kind: 'any'; of: readonly Condition[] }
  | { kind: 'not'; cond: Condition };

type Action =
  | { kind: 'setFlag'; flag: FlagId; value: boolean }
  | { kind: 'sendPuzzleEvent'; puzzle: PuzzleId; event: string; payload?: EntityI
  | { kind: 'giveItem'; item: ItemId }
  | { kind: 'takeItems'; item: ItemId | 'allCarried' }
  | { kind: 'startDialogue'; dialogue: DialogueId }
  | { kind: 'cue'; cue: CueId }                 // presentation-only feedback
  | { kind: 'checkpoint'; id: CheckpointId };

interface Interactable {
  id: EntityId;
  range: number;                  // metres from Wick's collider centre
  prompt: PromptId;
  targetableWhen: Condition;
  revealable?: EntityId;          // if set, the targeting system also requires i
  onInteract: readonly Action[];
  blockedWhen?: Condition;        // if true, play onBlocked instead
  onBlocked?: readonly Action[];  // must include a cue
}

interface Revealable {
```

```
  id: EntityId;
  mode: 'persistOnReveal' | 'whileLit';
  dwellMs: number;                // default 300
}

interface Transition<S extends string> { guard?: Condition; target: S; actions?:
interface PuzzleDefinition<S extends string, E extends string> {
  id: PuzzleId;
  initial: S;
  states: Record<S, {
    status: PuzzleStatus;
    transientMs?: number;         // required for transient states, ≤ 2000
    after?: S;                    // target state when transientMs elapses
    onEnter?: readonly Action[];
    on?: Partial<Record<E, readonly Transition<S>[]>>; // first matching guard wi
  }>;
  onReload: S | 'checkpoint';
  onRestart: S;
}

interface SaveDataV1 {
  schemaVersion: 1;
  levelId: 'graveyard';
  checkpointId: CheckpointId;
  elapsedPlayMs: number;
  flags: FlagId[];
  revealedIds: EntityId[];        // persistOnReveal only
  bonesDelivered: BoneId[];       // set semantics; carried bones are never saved
  completedPuzzleIds: PuzzleId[];
  collectedItemIds: ItemId[];     // unique collectibles
  bestCompletion?: { timeMs: number; collected: number; total: number };
}
```

Level data is validated at build time and in tests. Every referenced ID must exist, every puzzle event must be defined by its puzzle, and every interactable must emit at least one cue.

10.6 Source layout

```
little-haunts/
  public/assets/{characters,environments,props,audio,ui}/
  src/
    app/                # boot, loop, lifecycle, settings wiring
    engine/             # renderer, camera, assets, input, audio, persistence, pr
    game/               # PURE: state, step, collision, lantern, interactions, pu
      math/             # deterministic angle/vector helpers
```

```
  levels/graveyard/   # PURE data: puzzles, interactables, dialogue, beats, col
  shared/             # PURE types/utilities
  ui/                 # HUD, prompts, menus, completion
  styles/
tools/                # asset validation, collider/anchor extraction, manifest
spikes/               # disposable; excluded from build and from architecture r
tests/{unit,replay,graph,e2e}/
docs/{art-bible,architecture,asset-pipeline,level-design,spikes}/
asset-manifest.json
```

---

11. Art & Asset Production Pipeline

11.1 Process

1. Draft an original character sheet and environment concept board from registered references (§3). Finalize once the pending references arrive.
2. Lock scale, silhouette, palette, materials, and lighting standards.
3. Generate candidate models with Tripo or an equivalent tool. Do not assume they are game-ready.
4. In Blender: clean the topology, UVs, and materials; rig to the budget in §11.2; author clips; bake lighting for environments.
5. Export GLB files and validate them with `tools/` (orientation, scale, pivots, naming, clips, texture sizes, triangle counts).
6. Compress (Meshopt or Draco, KTX2) and generate lower-cost variants where needed.
7. Import into a test scene and check silhouette, animation, draw calls, texture memory, and frame time on the reference machine.
8. Accept into the asset library only after technical validation **and** human art review (§14.4).

11.2 Wick rig and asset budget (provisional; confirmed by the spike)

- **Height:** about 1.05 m in world units (≈0.6× a 1.75 m human, per REF-01), about 3.5 heads tall. The head, including the hood, is about 40% of the figure. Readability target in §14.2.
- **Bones (24 or fewer):** root, body/hover, spine, head, hood tip ×2, cloak (left/right/back, 1–2 each), arm and hand on each side with a lantern socket on the **left** hand, and tail ×4.

The mitten hands need no finger bones.

- **Face:** texture atlas or 10 or fewer morph targets, covering the expression list in §4.2.
- **Triangles and textures:** Wick 15k triangles or fewer; textures at most 2 × 1024² KTX2. The lantern is a separate GLB on the hand socket with an emissive material.
- **Felt material:** high roughness with optional sheen. The stitch seams are in the normal or albedo map, not the geometry. The fuzzy hood-edge halo is a shell or alpha fringe only if the spike shows it's affordable; otherwise it's baked into the textures. The spike measures the cost.
- **Ghost tail:** translucent pale-cyan material with a soft fresnel edge. It must stay readable against fog without becoming a bloom source.
- **Lantern:** a separate GLB, 3k triangles or fewer, with a modeled frame, a glass panel with emissive inner glow, and a top ring that attaches to the hand socket. The flicker is driven by a presentation-layer cue (deterministic seeded noise, so replays match). Its look follows REF-01's turnaround (front, side, back, top, bottom).
- A floating, cloaked character with no legs is the main simplification of v0.2. The spike validates it end to end before any level work (§12).

11.3 Level authoring conventions (Blender → data)

Prefix	Meaning	Extracted to
`COL_`	Static collider: box, rotated box, cylinder, curve, or zero-thickness fence line. Compiled to segment and circle primitives (§7.1).	`levels/<id>/collision.json`
`TRG_`	Trigger volume	`collision.json`
`SPN_`	Spawn point (Wick, NPC, bone)	Level anchors
`INT_`	Interactable or revealable anchor	Level anchors
`OCC_`	Occluder (fades per §5.1)	Render tags
Ground mesh	Height for presentation	Baked height grid (0.25 m cells)

Extraction runs at build time in Node, so the gameplay layer never parses GLB files. It fails on degenerate primitives, such as zero-length segments or non-positive radii. Level data refers to anchors by name, and validation fails the build if a referenced name is missing.

11.4 Rights and provenance

Maintain `asset-manifest.json` with an entry for every shipped file: owner, provenance (tool and model or author), license, source file, exported GLB, size, texture budget, and QA status. No third-party art or music without verified rights. AI-generated music and sound follow their service's actual license terms. CI fails if any file in `public/assets/` is missing from the manifest, has an `unknown` license, or isn't marked `accepted`.

---

12. Technical Spike (Phase 1a — disposable)

Purpose: find out whether the pipeline and runtime can deliver the REF-00/REF-01 look within the §13 budgets, and how much production will cost. **The spike validates feasibility only. It doesn't produce final assets.**

Timebox: 3 working days. Any overrun needs owner approval. By default, a question the spike can't answer is reported as a risk; the timebox isn't extended for it.

Kept separate from production

- **Spike assets** are feasibility-grade: good enough to answer a question (does it rig, does it render, what does it cost), not to pass the art rubric. Each is tagged `spike-only` in the manifest and can never be marked `accepted` or shipped.
- **Final assets** (Wick, the lantern, and the graveyard) are made in a separate production track that starts after Gate 1a (§16). That track follows §11 and uses the spike's measured findings.
- **Spike code** lives on branch `spike/wick-pipeline` under `spikes/`. It's never merged into `src/`, and it's excluded from the build and from the architecture rules. Production code may borrow ideas from it, not files.
- **Licensing rules** (§11.4) still apply to everything in the spike.

Plan

Day	Focus	Questions
1	Asset pipeline	**(1)** Can a Tripo-generated Wick be cleaned up, rigged to the §11.2 budget, and exported? Log hours per step so production effort can be extrapolated. **(2)** Do the cloak, mitten hands, and tail come out of cleanup clean enough to rig without remodeling? **(3)** Do Blender lightmaps display correctly in Three.js (second UV set, color space, tonemapping)?

2	Runtime	**(4)** Vignette performance for one graveyard corner with fog, particles, and post-processing: frame time p50/p95/p99, draw calls, triangles, texture memory, and transfer size, on the reference machine or a clearly labeled stand-in. **(5)** Lantern light: realtime shadowed point light versus shader-based radial light, with the frame-time cost of each. **(6)** Swept collision and slide around tombstones, fences, and corners: does it feel right, and does it pass the thin-geometry gauntlet (§14.1)? **(7)** Does occluder fade keep Wick visible behind foreground walls without looking cheap?
3	Look, motion, report	**(8)** Do procedural hover, sway, lean, and tail plus the clips read as expressive or just floaty? Provide recordings for owner review. **(9)** Face: texture atlas or morph targets? **(10)** What do the felt sheen and the hood-edge fiber halo cost compared with a roughness-only fallback? Write the report in the afternoon.

Deliverable: `docs/spikes/wick-pipeline.md` , containing:

- A findings table with measurements.
- Screenshots and recordings.
- The Blender hours log.
- Recommended changes to §4, §7, §10, §11, and §13.
- A production estimate with confidence ranges, split into engineering and asset production.

It’s reviewed at Gate 1a (§16).

---

13. Performance & Accessibility Requirements

13.1 Reference environment

- The **reference desktop** (CPU, GPU, RAM, OS) and **reference browser** must be fixed before Phase 1b begins (D4).
- A **minimum-spec** integrated-GPU laptop is fixed at the same time.
- All performance numbers are measured with the in-game perf harness (`?perf=1`), which logs per-frame times, draw calls, triangles, and an estimate of texture memory to a downloadable JSON file. Each run follows a scripted 120 s route driven by replayed input, with the browser in focus.

13.2 Budgets (provisional; revised once after the spike, then fixed at Gate 1b)

Metric	Target
Frame time, reference desktop, 1920×1080, High preset	p50 ≤ 16.7 ms, p95 ≤ 20 ms, p99 ≤ 25 ms
Frame time, minimum spec, Low preset	p50 ≤ 33.3 ms
Hitches after the Ready screen	No frame > 100 ms (shaders are precompiled during loading)
Compressed transfer to first interactive frame	≤ 25 MB
Total level transfer including lazy-loaded optional content	≤ 40 MB
JavaScript (gzipped)	≤ 600 KB
Time to interactive, cold cache, 50 Mbps throttled, reference desktop	≤ 10 s
Steady-state draw calls	≤ 200
Visible triangles	≤ 750k
Estimated GPU texture memory	≤ 384 MB

Quality presets: Low, Medium, and High control shadow quality, post-processing, particle density, and render scale.

13.3 Robustness

- A loading progress indicator, plus a friendly fallback page when WebGL2 is unavailable.
- **Lost WebGL context** (simulated with `WEBGL_lose_context`): the game restores and resumes, or shows a recoverable error. Never an uncaught exception.
- **Tab hidden** (`visibilitychange`): the game auto-pauses, saves elapsed time, and stops accruing play time.
- **Audio:** none created or resumed before the first user gesture.

13.4 Accessibility

- All UI is operable by keyboard, with visible focus indicators.

- HUD, prompt, and dialogue text meets a contrast ratio of at least 4.5:1 (WCAG AA) against its background panel.
- **Reduced motion setting:** hover bob amplitude drops to 25% or less, there's no camera shake, scripted camera moves become cuts, and full-screen flashes are disabled.
- No content flashes more than 3 times per second (WCAG 2.3.1).
- Separate master, music, and sound-effects volume controls; the lantern can be set to hold or toggle.
- Prompts are at least 18 px tall at 1080p, and they scale with the viewport.

---

14. Acceptance Criteria

There are four kinds of criteria. A gate passes only when every criterion assigned to it passes.

14.1 Automated (CI, on every PR)

1. Typecheck (both tsconfigs), lint, and dependency-cruiser pass.
2. Unit tests cover **every transition** in §9.1–9.3 and every row of §9.4. A coverage report maps each test to its transition ID.
3. The replay determinism suite passes (§10.4).
4. The soft-lock graph search (§9.5) and the geometric reachability test pass.
5. Level data validation passes: all IDs resolve, every interactable emits a cue, transient states are 2000 ms or shorter, and every checkpoint is reachable. At least three distinct interactable types each resolve to a visual cue **and** an audio cue.
6. Save codec tests pass: round trip, migration, and corrupted, unknown, or future data falling back safely with a backup written.
7. **End-to-end smoke tests** (Playwright Chromium, Firefox, WebKit): load to the Ready screen; movement with the keyboard; pausing and resuming; reloading at each checkpoint restores the expected state; a scripted playthrough (replayed intents through the real renderer) reaches the completion screen showing time and collectible count; zero console errors.
8. Audio stays suspended until the first input, checked through the AudioContext state.
9. Bundle and transfer budgets in §13.2 are checked against the production build.

10. The asset manifest check passes (§11.4).
11. The owner's holdout acceptance tests pass.
12. **Thin-geometry collision gauntlet:** Wick is swept into zero-thickness segments, boxes, and corners from every heading at 1° steps, at speeds from 0 up to 10× the maximum dash speed. That includes single-tick moves longer than his collider. He never crosses or ends up overlapping any primitive, slides along walls, and stops cleanly in acute corners. Fast dashes across the exit threshold always fire its trigger.
13. **Reveal-gated interaction:** for each `whileLit` interactable, lowering the lantern or leaving the light removes the prompt and the target once the grace period ends, and an interact press on any later tick has no effect. Unrevealed `persistOnReveal` objects are never targetable.
14. **Bone conservation:** the §9.2.1 property tests and invariant checks pass.

14.2 Measured on hardware (at each gate from 1b on)

1. The §13.2 frame-time, hitch, load, draw-call, triangle, and memory budgets are met on the reference and minimum-spec machines. Perf harness JSON files are attached to the gate review.
2. **Wick readability:** at default framing on 1080p, Wick is at least 64 px tall on screen. In a grayscale capture at 25% scale, he is identifiable by silhouette in all canonical shots (§14.4).
3. Context loss and tab suspension behave as specified in §13.3.
4. Manual pass on current stable desktop Chrome, Edge, Firefox, and Safari: the full playthrough completes in each with no broken animations, missing textures, invisible walls, or soft-locks.

14.3 Structured playtest (Gate 3 and Gate 4)

At least 5 first-time testers, given no verbal instruction, using the reference or a comparable desktop. Each session is screen-recorded, and stall points are logged.

Metric	Pass threshold
Testers who reveal the first rune within 3 minutes of gaining control	≥ 4 of 5
Testers who complete the main path without help	≥ 4 of 5
Median first-playthrough time	5–10 minutes

Testers who discover the skeleton gag	5 of 5 (it's on the main path)
Testers who find the secret	≥ 1 of 5 (it should be findable, but not obvious)

Results go in `docs/playtests/`. A failed threshold produces a fix list before the gate is retried.

14.4 Human art review (owner sign-off)

Artistic quality stays a human judgment, applied consistently:

- **Canonical shots:** 8 fixed camera and pose setups, defined in Phase 0 and captured at the same settings every time. They include Wick idle, Wick lantern raised, the lamp circle lit, the skeleton scattered, the courtyard reassembly, the secret alcove, the exit gate, and a wide establishing shot.
- **Rubric** (scored 1–5 per shot): style match to the registered references; Wick's appeal and expressiveness; premium diorama material quality; lighting mood and warm/cool contrast; atmosphere (fog, particles, depth); readability of paths and interactables; consistency across assets.
- **Pass:** every dimension scores 4 or higher on every shot, plus owner approval of the motion recordings showing hover, sway, lean, the expression set, and the skeleton gag.
- The art bible can't be finalized, and Gate 2 can't pass, until the owner's final approval of REF-00 and REF-01 is recorded (§3).

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15. Definition of Done (MVP)

- Every §14.1 check is green on `main`.
- Every §14.2 measurement passes on both machines, with artifacts attached.
- The §14.3 playtest thresholds are met.
- The §14.4 art review is signed off.
- 100% of shipped assets have manifest entries with verified rights.
- The game is deployed to Cloudflare Pages at a public URL, and the deployed build matches the build that was tested (same commit hash, shown in the pause menu).

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16. Milestones & Approval Gates

Phase	Deliverable	Gate (owner approves)
0. Preproduction	Updated GDD, draft art bible and asset checklist, provisional architecture, level layout and state diagrams checked against §9, spike plan, risk register, canonical-shot definitions	Scope, art direction, and spike plan approved. **The spike may not start before this gate.**
1a. Spike (3 days, disposable)	Spike report (§12); feasibility-grade assets only	Pipeline go/no-go; architecture, budgets, and estimate finalized; production asset track authorized
1b. Technical proof	Production scaffold; CI with all §10.4 checks live; camera, movement, collision, lantern reveal, interaction targeting; Wick GLB with procedural motion	§14.1 items 1, 3, 7 (movement), 12, and 13 pass; mechanics feel approved; §14.2 item 1 baseline captured
2. Visual proof	One polished graveyard vignette with final lighting and the final Wick	REF-00/REF-01 final approval recorded (§3); §14.4 pass on the shots this vignette covers; §14.2 items 1–2 pass in the vignette
3. Vertical slice	Both puzzles, the NPC, the secret, saves, UI, audio, and the completion screen	All of §14.1; §14.2 items 1–3; §14.3
4. Release polish	Browser compatibility, optimization, deployment	All of §15

Effort: not committed until Gate 1a. The spike report supplies the estimate with confidence ranges.

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17. Principal Risks & Mitigations

Risk	Mitigation
AI-generated 3D models	Floating, legless Wick removes leg rigging and foot contact; the

are inconsistent or hard to rig	spike validates the full model-to-browser pipeline before any level work
Procedural motion feels floaty or lifeless rather than expressive	Spike recordings reviewed by the owner; all motion parameters tunable in config; authored clips carry the key emotional beats
Beautiful screenshots but poor runtime performance	Spike vignette profiled on reference hardware; budgets enforced at every gate from 1b on
Visual mismatch across generated assets	Reference register, art bible, standardized Blender materials and export, rubric-based review
Missing or unapproved references lead to guessed art direction	Agents may not substitute references; the art bible stays draft and no asset is accepted until REF-00 and REF-01 have the owner's final approval
Custom collision feels sticky or leaks through thin props	Swept collision with slide, tunable radius, debug overlay, thin-geometry gauntlet, spike Day 2 validation; colliders authored smoother than the visible mesh
Foreground diorama geometry hides Wick or puzzle elements	Occluder fade (§5.1), scripted framing rules, canonical-shot review
Determinism quietly erodes through builder agents	Mechanical enforcement (§10.4), cross-engine replay hashes, holdout tests
Puzzle soft-locks	Normative state machines, transition coverage, graph search, and geometric reachability tests
Scope grows into a full game	Explicit non-goals (§2), future content listed separately (§4.5), approval gates
Browser input or audio differences	Intent abstraction, gesture-gated audio, three-engine end-to-end tests, manual Safari pass
Unclear asset licensing	Manifest with CI enforcement (§11.4)

18. Reusability for Mostly Human Studio

Keep the rendering, input-intent, collision, audio, persistence, and asset-tooling systems reusable. Keep levels, puzzle definitions, characters, and art packs replaceable. The

deterministic core, data contracts, and graph-based soft-lock checks are likely the most portable pieces. Do **not** build a generic game factory before Little Haunts proves a complete, polished slice. The first reusable production capability to document is the asset intake and QA pipeline (§11), including the spike's measured findings.

19. Agent Assignment (Planning Only)

Act as lead game engineer, game designer, and technical artist. Review this specification and the references registered in §3. Produce the following:

1. An updated GDD, including the final rune-order encoding and dialogue drafts.
2. A provisional technical architecture with tradeoffs, consistent with §10, including any points that depend on the spike's outcome.
3. A draft art bible and asset checklist, labeled draft until the owner gives final approval of REF-00 and REF-01.
4. A spike plan: steps, measurements, tools, the 3-day timebox and the separation from production assets, and how each §12 question gets answered.
5. A graveyard layout plus puzzle and NPC state diagrams. Validate them against §9 and list any deviations, each with its rationale.
6. Implementation milestones and a task breakdown that follows §16.
7. Confirmation of, or proposed changes to, the §13–14 numbers, each with its rationale.
8. A list of unresolved decisions, including any new ones.

Constraints

- Do not write production code. Do not start the spike until the owner approves Gate 0.
- Do not generate, search for, or substitute visual references. Use only the §3 register, and report gaps as blocking questions.
- Recommend changes with rationale. Do not apply changes to this specification without owner approval.

20. Decisions

Closed in v0.2

#	Decision	Resolution

D1	What Wick is	A floating ghost: cloak and tail, no legs (§4.1)
D2	Camera	Fixed isometric; no rotation in the MVP (§5.1)
D3	Input	Keyboard-only MVP; the intent layer allows click-to-move and gamepad later (§5.3)
D7	Physics	No physics engine in the MVP; custom 2D collision (§7.1)

Open

#	Decision	Needed by
D4	Reference desktop, minimum-spec laptop, and reference browser	Before Phase 1b
D5	Whether the first public demo needs mobile controls or only a responsive layout	Gate 3
D6	Music production method and commercial rights verification	Gate 3
D8	Final camera pitch and framing size	Gate 2
D9	Rune-order encoding (notches or an alternative)	Gate 0 (GDD)
D10	Lantern lighting technique (realtime shadowed or shader radial)	Gate 1a
D11	Facial expression technique (atlas or morph targets)	Gate 1a
D12	Felt sheen material, or roughness-only fallback	Gate 1a
D13	Owner final approval of REF-00 and REF-01 as primary art references, including confirming the lantern hand	Before Gate 2

Approval status: Direction approved at the v0.2 review. v0.3 awaits owner confirmation. No implementation started. **Success criterion:** A delightful, cohesive, original 3D game people can play directly in their browser, with that delight checked by measured criteria and owner review.